

Exercise 2 – REACT NATIVE MESSAGES DIRECTORY

Project Title: Message Box – Directory Based Message App

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Project Overview

- **Messages Directory App** is a simple React Native mobile application developed using **Expo**. The main purpose of this app is to organize different types of messages into separate directories such as Home, Family, Friends, Love, University, and You.
- The app shows a clean dark-themed user interface. On the home screen, users can see different message categories in a grid format. When the user taps on any directory, the app opens a message screen and displays the related messages for that category.

Tools and Software Required

To run this project, the following tools are required:

- Node.js
- npm
- Expo CLI
- React Native
- Android Studio or Expo Go mobile app
- Visual Studio Code or any code editor

Technologies Used

- React Native
- Expo
- JavaScript
- React Navigation
- Native Stack Navigation
- FlatList
- SafeAreaView
- StyleSheet

Main Features

- Home screen with message directory cards
- Dark theme user interface
- Grid layout for categories
- Category-wise message display
- Separate message screen for each directory
- Numbered message cards
- Special achievement layout for the **You** section
- Custom colors for every directory
- Simple navigation between screens

Project Structure (Important Files)

```
MessagesDirectoryApp/  
|  
├─ App.js  
├─ app.json  
├─ babel.config.js  
├─ package.json  
|  
├─ assets/  
|   ├─ app-icon.png  
|   ├─ family.png  
|   ├─ friends.png  
|   ├─ home.png  
|   ├─ love.png  
|   ├─ school.png  
|   └─ you.png  
|  
├─ data/  
|   └─ messages.js  
|  
└─ screens/  
    ├─ HomeScreen.js  
    └─ MessageScreen.js
```

Important File Description

- **App.js**

This is the main entry file of the application. It sets up the navigation container and creates two screens:

- Home Screen
- Messages Screen

It also applies a dark theme for the application.

- **HomeScreen.js**

This screen displays all message directories in a two-column grid layout. Each directory card contains an initial letter and title. When the user taps a card, it navigates to the message screen.

- **MessageScreen.js**

This screen displays messages based on the selected directory. Normal directories show simple numbered messages, while the You section shows achievement-style cards.

- **messages.js**

This file stores all directory data, including category names, colors, descriptions, and messages.

- **package.json**

This file contains the project dependencies and scripts required to run the application.

Features Implemented

The MessageNest – React Native Messages Directory app includes the following features:

1. **Directory-Based Home Screen**

The home screen displays different message directories such as Home, Family, Friends, Love, University, and You.

2. **Grid Layout Design**

All directories are shown in a clean two-column grid layout, which makes the app easy to use and visually organized.

3. **Category-Wise Messages**

Each directory contains its own related messages. When the user selects a directory, only messages from that selected category are displayed.

4. **Screen Navigation**

React Navigation is used to move from the home screen to the message screen smoothly.

5. **Dynamic Data Display**

The app stores message data in a separate data file and displays it dynamically using React Native components.

6. **Custom Directory Styling**

Each directory has its own color, icon/emoji, title, and description to make every section easy to identify.

7. **Message Cards**

Messages are displayed in card format, which gives the app a clean and readable look.

8. **Dark Theme UI**

The app uses a modern dark theme design for a better user experience.

9. **Special “You” Section**

The You directory contains personal motivational and achievement-style messages with a different layout.

10. **Simple and User-Friendly Interface**

The app is designed with simple buttons, readable text, and easy navigation, making it suitable for all users.

Outputs :

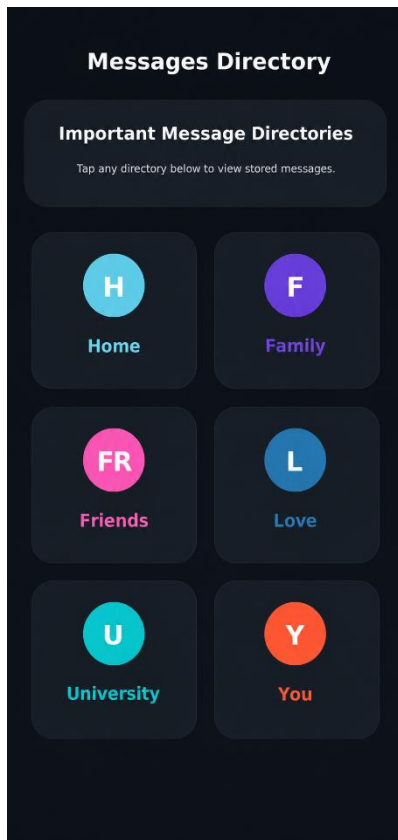


Figure 1: Home Screen

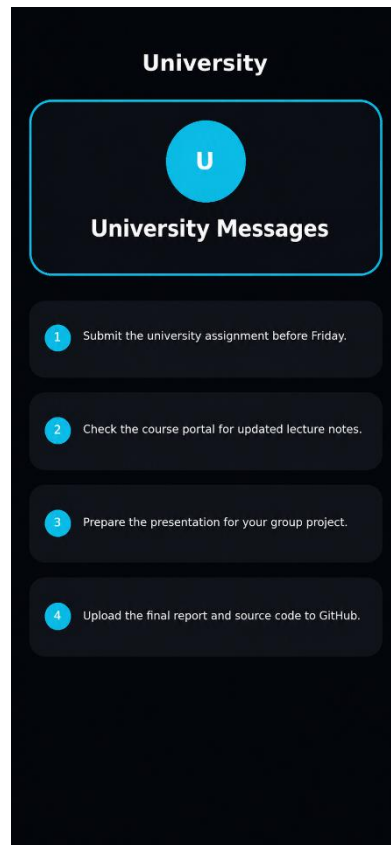


Figure 2: University



Figure 3: Family

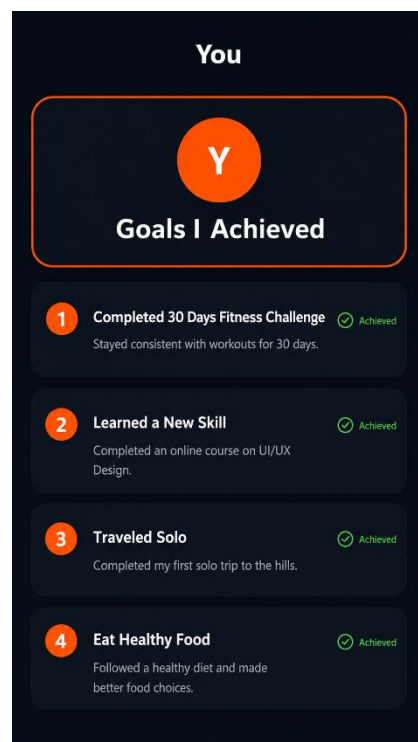


Figure 4: You

Challenges Faced

- During this project, I faced challenges while setting up the React Native and Expo environment because all dependencies needed to be installed correctly. I also had some issues with navigation, especially passing data from the home screen to the message screen. A small mistake in the import path or parameter name can create errors and stop the app from running properly.
- Another challenge was designing the directory cards in a clean grid layout and showing category-wise messages based on the selected directory. I had to make sure that each category displayed only its related messages with proper styling. These challenges helped me understand React Native project structure, navigation, dynamic data handling, and UI design more clearly.

Future Enhancements and Results Summary

- In the future, this app can be improved by adding features like adding new messages, editing messages, deleting messages, copying messages, and saving favorite messages. Local storage can also be added so that saved messages remain available even after closing the app. A search option, light/dark mode switch, and better icons or images can also make the app more useful and attractive.
- The final result of this project is a working React Native messages directory app with multiple categories and category-wise message display. The app successfully shows directories on the home screen, allows users to open each category, and displays related messages in a clean card format. Overall, this project helped me practice React Native basics, navigation, data handling, and mobile UI design.

Conclusion

The **MessageNest – React Native Messages Directory** app was successfully developed using React Native and Expo. This project helped me understand how to create a mobile app with multiple screens, use navigation, pass data between screens, and display category-wise messages in a clean layout. Overall, this exercise improved my practical knowledge of React Native, component-based design, styling, and dynamic data handling. The final app is simple, user-friendly, and useful for organizing messages into different directories.

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